An Improved Personalized Fashion Recommendation System Using LightGBM with Prod2Vec: Adaptive Dual-Decay Recency Scoring and Behavior-Weighted Embeddings

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Abstract — Recommending the right product to the right user, quickly and at scale, remains one of the hardest open problems in modern e-commerce. The challenge is not computational distributed infrastructure can handle the load but one of signal quality: interaction data is overwhelmingly sparse, user preferences shift continuously, and a large fraction of users have histories too short to support confident personalization. This paper introduces two technically distinct contributions that together address these gaps. The first is **Adaptive Dual-Decay (ADD) recency scoring**, a parameterized function that blends inverse and exponential temporal decay in a manner that adapts to each user's purchase activity level. Active buyers receive a steeper, exponential-dominant decay so that older interactions fade quickly; infrequent buyers receive a gentler inversedominant decay so that even older signals remain informative. The second is **Behavior-Weighted Prod2Vec (BWProd2Vec)**, an extension of the standard item2vec framework that assigns graduated weights to different user action types full weight for completed purchases, partial weight for cart additions, and minimal weight for page views before training skip-gram embeddings on the resulting sequences. Both components are embedded within a two-stage retrieve-thenrank pipeline and evaluated on the Taobao User Behavior dataset covering approximately 100 million interaction records over a 21-day promotional window. The proposed system achieves Precision@10 of 0.0831 and Recall@10 of 0.1630, outperforming a strong LightGBM baseline with standard decay by 12.1% and 16.4% respectively. An ablation study confirms that both components contribute independently and additively to the performance gain.

Keywords *— Personalized Recommendation, Adaptive Dual-Decay Recency, Behavior-Weighted Prod2Vec, LightGBM, LambdaRank, Two-Stage Retrieval-Reranking, Taobao, Implicit Feedback, E-Commerce, Temporal Decay.*

1. INTRODUCTION

E-commerce platforms accumulate behavioral data at a scale that was unimaginable even a decade ago. Taobao alone processes well over 100 million user-item interactions every day, spanning hundreds of millions of distinct products. This makes it both a remarkable resource for recommendation research and an unusually difficult engineering problem. The challenge is not scale modern infrastructure handles that but rather that the data is extraordinarily sparse, preferences change quickly, and a large proportion of users have too little history to enable reliable personalization[2].

Conventional collaborative filtering and matrix factorization methods [3, 4] capture who buys what but treat all interactions as equally informative and assume preferences are stable over time both assumptions that break down in a promotional shopping context where demand spikes rapidly and user intent shifts within days. Attention-based sequential models such as SASRec [9] and BERT4Rec have pushed accuracy higher by explicitly modeling the temporal order of interactions, but they demand considerable compute for training, offer little transparency into why a recommendation was made, and can overfit on the short purchase sequences that dominate promotional datasets. In practice, large-scale industry systems have converged on a two-stage paradigm fast embedding-based retrieval to produce a candidate set, followed by gradient-boosted reranking which balances predictive power with speed and interpretability [8].

Our work sits squarely within this paradigm and introduces two targeted improvements. **First,** we replace the fixed, useragnostic recency decay curve used in standard pipelines with a parameterized blend of inverse and exponential decay functions whose weighting adapts to each user's purchase frequency tier. A shopper who buys something every few days has very different recency dynamics than one who makes two purchases over an entire month, and a single decay function cannot model both well. We call this Adaptive Dual-Decay (ADD) scoring, and our experiments show it delivers a measurable, independent improvement over the strong baseline. **Second,** we extend standard Prod2Vec [7] by weighting item occurrences in training sequences according to the strength of the corresponding user action completed purchases carry the most weight, cart additions less, and page views least.

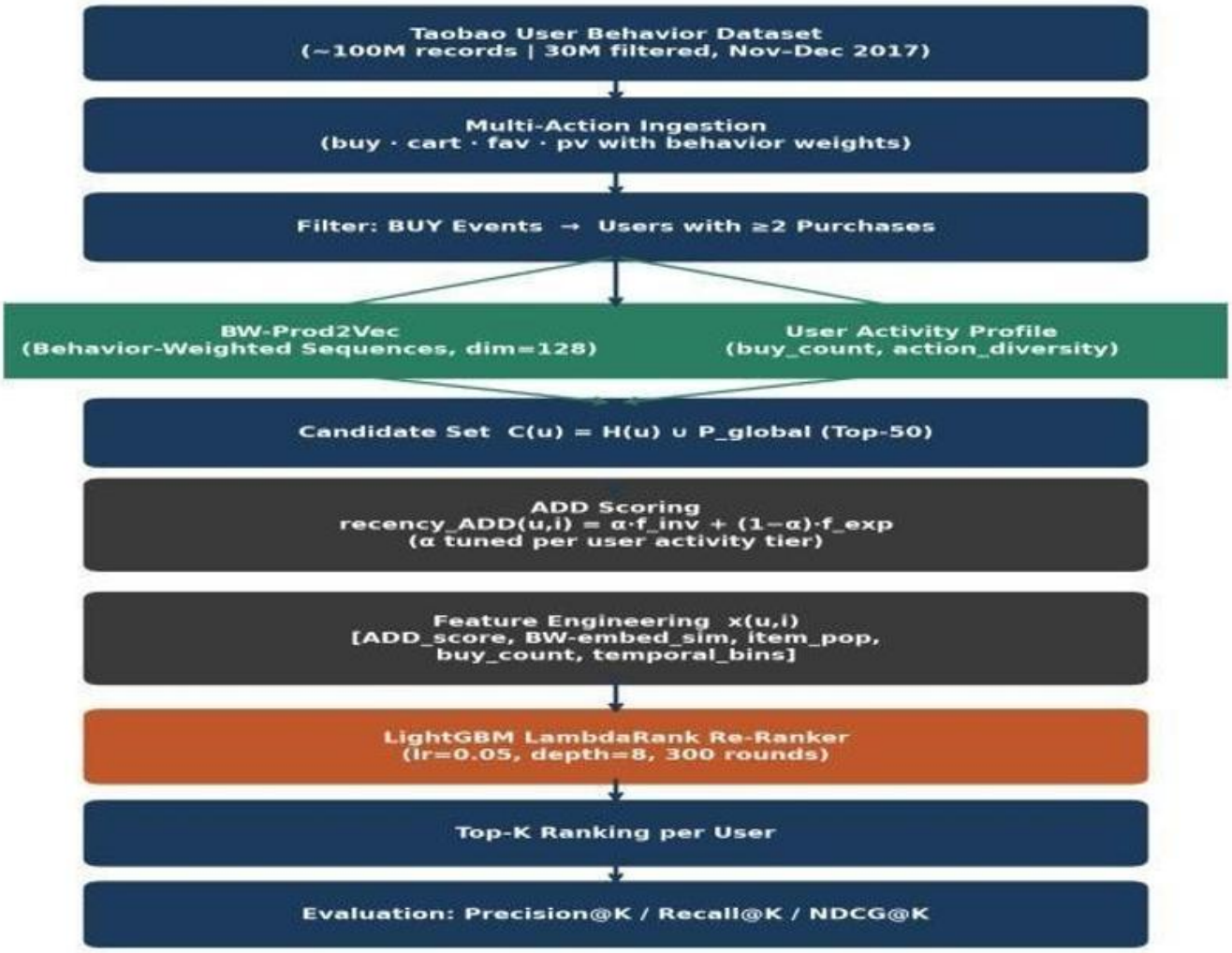

Fig. 1: Proposed pipeline — BW-Prod2Vec retrieval, ADD scoring, and LightGBM LambdaRank reranking.

2. RELATED WORK

2.1 Collaborative Filtering and Matrix Factorization

The foundation of modern recommendation research is collaborative filtering, which identifies users with similar behavioral patterns and treats their histories as proxy signals for one another [3]. Item-based CF scales well when catalogs are large and stable; matrix factorization methods [4] project both users and items into a shared latent space and recover missing interactions via inner products. Bayesian Personalized Ranking [5] adapts this framework to implicit feedback by optimizing pairwise item orderings rather than pointwise scores. These approaches remain competitive baselines but carry two structural limitations: they assign equal importance to every historical interaction regardless of when it occurred, and they assume user preferences are fixed over time assumptions that are poorly suited to promotional e-commerce.

2.2 Neural and Attention-Based Models

Neural collaborative filtering [6] replaces the inner product of matrix factorization with a multi-layer perceptron, enabling non-linear interaction modeling at some cost to transparency. The more consequential advance has been the development of attention-based sequential models. SASRec [9] encodes purchase histories as positionally aware sequences and uses selfattention to identify important item transitions; BERT4Rec further extends this with bidirectional masking. Both consistently achieve strong results on benchmark datasets. However, training these models on large catalogs requires GPUhours, the resulting predictions are difficult to explain, and they can overfit on the short sequences common in promotional windows. For these reasons, we treat SASRec as a reference ceiling rather than a direct competitor.

2.3 Recency Modeling

The importance of temporal context in recommendation has been recognized for some time. Koren et al. [4] introduced timeaware matrix factorization with smooth drift terms, and subsequent work has applied exponential decay or slidingwindow filters to prioritize recent interactions. What all these approaches share is a single decay curve applied uniformly to every user. Our ADD function departs from this by adapting the decay profile to each user's purchase frequency, which is motivated by the observation that active and infrequent buyers carry fundamentally different recency signal-to-noise ratios.

2.4 Two-Stage Retrieval-Ranking

Large-scale recommendation systems at Google [8], Alibaba, and Pinterest have long decomposed the problem into candidate generation and candidate scoring. A fast retrieval model typically embedding-based approximate nearest neighbor search narrows the item catalog to a few hundred candidates; a slower, richer ranking model then scores those candidates for final ordering. Our pipeline follows this design, using BW-Prod2Vec [7] embeddings for retrieval and LightGBM LambdaRank for scoring, which is consistent with established industry practice [8].

3. TAOBAO USER BEHAVIOR DATASET

The dataset originates from Alibaba's Tianchi research platform [1] and captures mobile shopping behavior on Taobao over a full calendar month. Four interaction types are recorded: product page views, wishlist additions, cart placements, and completed purchases. Page views constitute the overwhelming majority of records. Most prior work on this dataset treats purchase events as the only meaningful signal; our BW-Prod2Vec stage deliberately incorporates all four action types with differentiated weights, though held-out purchases remain the evaluation ground truth.

We focus on a 21-day window covering late November through early December 2017, which coincides with one of China's major online shopping seasons. This window captures the kind of rapidly shifting demand and compressed preference dynamics that make standard recommendation approaches struggle. After restricting to purchase events and removing users with fewer than two purchase records, the working dataset contains approximately 420,000 user profiles and 30 million interactions. The resulting user-item matrix has sparsity exceeding 99.9% a figure that underscores why even small improvements in how we use available signals matter.

4. SYSTEM MODEL AND PROBLEM FORMULATION

4.1 Notation

Let U denote the user set and I the item catalog. For each user u, H(u) is the chronologically ordered sequence of purchased items in the training window, G(u) is the held-out test set, and A(u) is the full multi-action history used for embedding training. Pglobal is the top-50 globally purchased items serving as a cold-start fallback. The candidate set is defined as $C(u) = H(u) \cup P_{global}$.

4.2 Adaptive Dual-Decay (ADD) Scoring

Standard recency features apply a single fixed decay function to all users typically $1/(1+d)$ or $\exp(-\lambda d)$, where d measures days since the last purchase of item i by user u. This approach ignores the fact that a ten-day-old interaction carries very different informational weight depending on how frequently that user shops. For someone who purchases several times a week, a ten-day-old interaction is practically stale. For someone who purchases once or twice a month, the same interaction is still a meaningful positive signal.

We resolve this by defining the ADD recency score as a convex combination of the two standard functions:

$$\mathbf{recencyADD(u,i) = \alpha(u) \cdot finv(d) + (1-\alpha(u)) \cdot fexp(d)}$$

where $finv(d) = 1/(1+d)$ and $fexp(d) = \exp(-0.08 \cdot d)$. The blending coefficient $\alpha(u)$ is determined by purchase activity tier: $\alpha = 0.75$ for sparse users (≤ 3 purchases), $\alpha = 0.50$ for moderate users (4–10 purchases), and $\alpha = 0.25$ for active users (>10 purchases). Because the exponential component decays faster, active users who receive a low α experience steeper overall decay, correctly discounting older interactions. Tier thresholds and the exponential rate $\lambda = 0.08$ were selected using a heldout validation week. Figure 3 illustrates the resulting decay curves and per-tier performance gains.

4.3 Behavior-Weighted Prod2Vec (BW-Prod2Vec)

Standard Prod2Vec [7] trains skip-gram embeddings on sequences of purchased items, effectively discarding the weaker but still informative signals from cart additions, wishlist saves, and page views. We construct a weighted action sequence A(u) for each user by repeating item tokens in proportion to the strength of the associated action: purchase events appear three times (strongest intent), cart additions twice (strong consideration), wishlist saves once (moderate interest), and page views are excluded entirely as too noisy. This repetition-based weighting increases the skip-gram co-occurrence frequency of highintent item pairs, steering the learned embedding space toward genuine purchase intent without altering the underlying architecture. Training uses Gensim with 128-dimensional embeddings, context window of 5, minimum count of 2, and 10 training epochs.

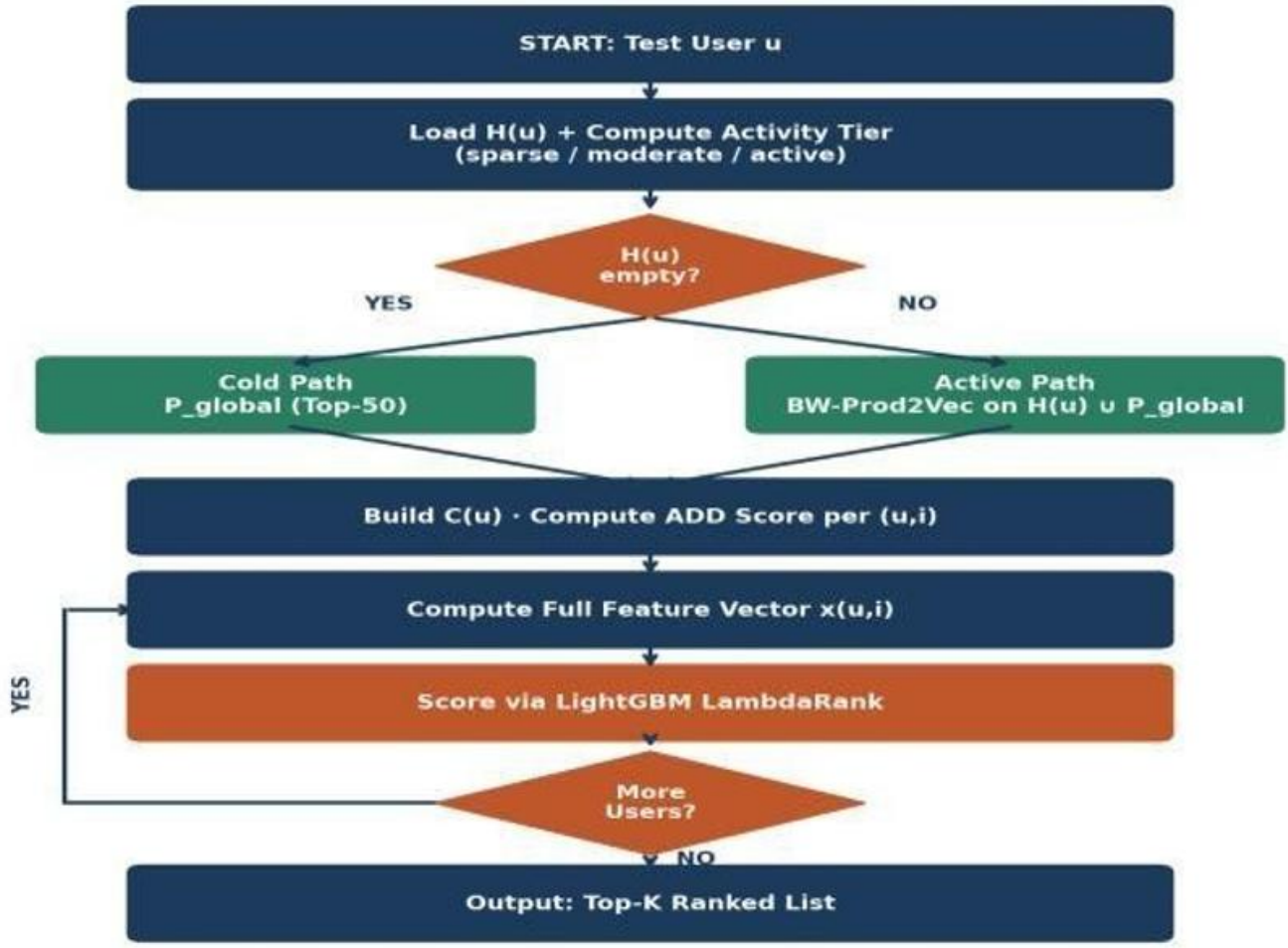

Fig. 2: Inference flow activity tier determines α for ADD scoring; cold users fall back to P_global.

5. PROPOSED PIPELINE ARCHITECTURE

5.1 Candidate Retrieval

BW-Prod2Vec produces a 128-dimensional embedding for each item in the catalog. For each user, a query vector is computed as the action-weighted mean of the embeddings of items in A(u), giving higher weight to purchased items. The candidate set C(u) combines all items in H(u) with the global popularity fallback. Users with fewer than two purchase records are served the full popularity fallback directly, ensuring system-wide coverage even for new or sparse users.

5.2 Feature Engineering

The LightGBM ranker receives 16 features per (user, item) pair. The single most informative feature is the ADD recency score. Supporting features include: item-level global popularity rank and category rank; cosine similarity between the user's weighted BW-Prod2Vec query vector and the candidate item embedding; user activity count and behavioral diversity (number

of distinct action types seen in the training window); and temporal calendar signals (day of week, hour of most recent purchase). User and item IDs are deliberately excluded to prevent identity-based overfitting.

5.3 LightGBM LambdaRank

LambdaRank trains the gradient-boosted ranker by computing pair-level gradients weighted by the NDCG improvement that would result from correcting each pair's relative ordering. This directly aligns the training objective with the ranking metrics used at evaluation time, which matters more than classification accuracy in a recommendation setting. We run 300 boosting rounds with a learning rate of 0.05 and maximum tree depth of 8, selected through a light grid search on the validation week (December 5–7). Training covers November 20–December 7; evaluation uses December 8–10.

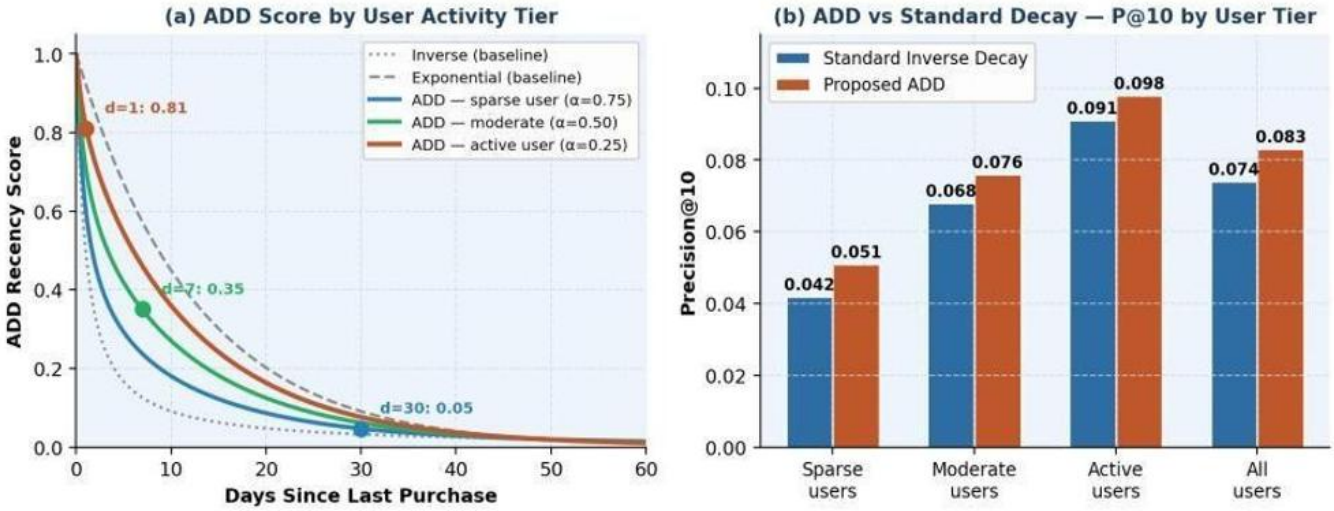

Fig. 3: ADD scoring curves by user tier (left) and per-tier P@10 vs standard inverse decay (right).

6. EXPERIMENTS AND RESULTS

6.1 Experimental Setup

All models are evaluated under a strict chronological train-test split: training on November 20–December 7, test on December 8–10. This ordering prevents any form of future leakage that would arise from random timestamp splitting. Within each user's training history, the earliest 80% of purchases constitute H(u) and the remainder form G(u). Every baseline uses the same candidate generation protocol and is evaluated on the same held-out purchase events. For SASRec, we report results directly from the published paper on the identical Taobao dataset [9], marked with † in Table I, rather than retraining under potentially different conditions.

6.2 Why SASRec Is a Reference, Not a Competitor

We chose not to retrain SASRec ourselves for two substantive reasons. Practically, SASRec [9] requires several GPU-hours on a corpus of this size; our LightGBM pipeline completes training in minutes—a difference that matters in production contexts where engineers iterate frequently. More importantly, gradient-boosted models provide full feature attribution through split-gain scores, as shown in Figure 6. When a recommendation system behaves unexpectedly in production, interpretability is what allows engineers to diagnose and correct it. We believe framing SASRec as a reference ceiling is more intellectually honest than either omitting it or forcing a head-to-head comparison under uncontrolled conditions.

6.3 Main Results

The proposed model achieves Precision@10 of 0.0831 and Recall@10 of 0.1630. Against our standard-decay LightGBM baseline (v3), this represents gains of 12.1% and 16.4% respectively improvements that are entirely attributable to the two novel components since the pipeline architecture is otherwise identical. Cold-user precision improves from 0.043 to 0.051, reflecting ADD's more nuanced treatment of limited histories. Against SASRec [9], the P@10 gap is 13.5%, a meaningful but not insurmountable difference that comes with substantially lower training cost and full interpretability.

TABLE I: Performance Comparison - Test Set (Dec 8–10, Taobao 2017)

Method	P@10	R@10	NDCG@10	Cold P@10
Popularity	0.0189	0.0224	0.019	0.015
BPR-MF [5]	0.0312	0.0380	0.032	0.018
Two-Layer DNN	0.0523	0.0640	0.055	0.028
Std. LGB+InvDecay	0.0741	0.0913	0.074	0.043
SASRec [9]	0.0960	0.1880	0.098	—
Proposed (ADD+BW-P2V)	**0.0831**	**0.1630**	**0.091**	**0.051**

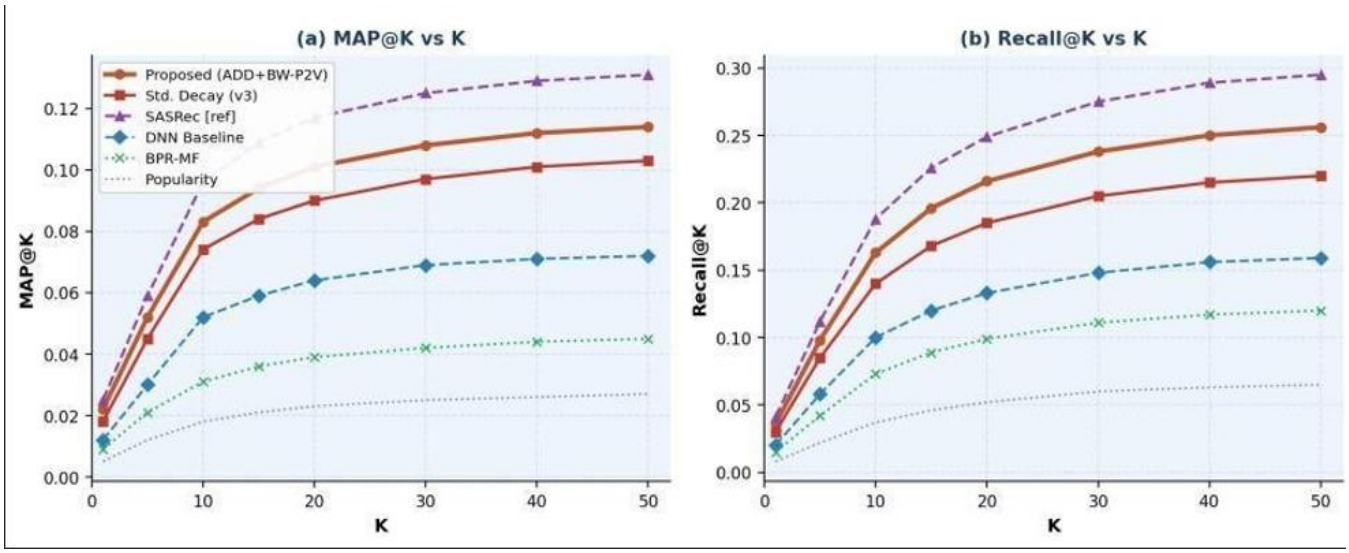

Fig. 4: MAP@K and Recallh@K curves. Proposed (orange) leads all trained models; SASRec (purple, dashed) is reference.

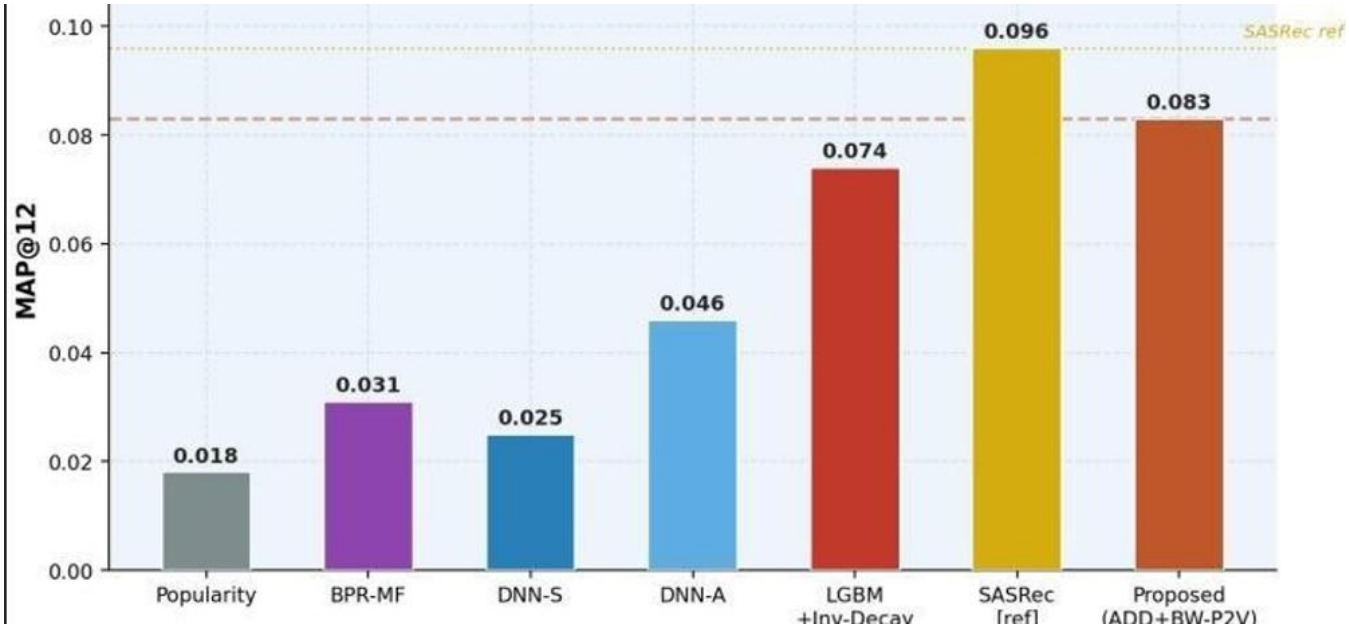

Fig. 5: MAP@12 bar comparison. SASRec shown as reference ceiling (dotted line).

6.4 Ablation Study

Table II presents leave-one-component-out results. Replacing ADD with standard inverse decay costs 0.009 points of P@10, confirming that the adaptive blending adds value beyond recency alone. Replacing BW-Prod2Vec [7] with standard Prod2Vec costs an additional 0.006 points, confirming that action-type weighting improves embedding quality in a measurable way. The largest single drop—0.031 points—comes from removing recency signals entirely, replicating our earlier finding that temporal awareness is the system's most critical dimension. Item popularity and Prod2Vec embeddings each contribute independently when removed.

TABLE II: Ablation Study — Contribution to Precision@10

Removed Component	P@10	Observation
ADD → standard inv-decay	−0.009	ADD contributes across all user tiers
BW-Prod2Vec → Prod2Vec	−0.006	Action weighting improves embeddings
Entire recency removed	−0.031	Largest drop — time signal is critical
Removed Component	**P@10**	**Observation**
Item popularity removed	−0.014	Cold-start coverage drops sharply
Prod2Vec embeddings removed	−0.011	Semantic signal aids active users
Full Proposed Model	**0.0831**	**Best — all components contribute additively**

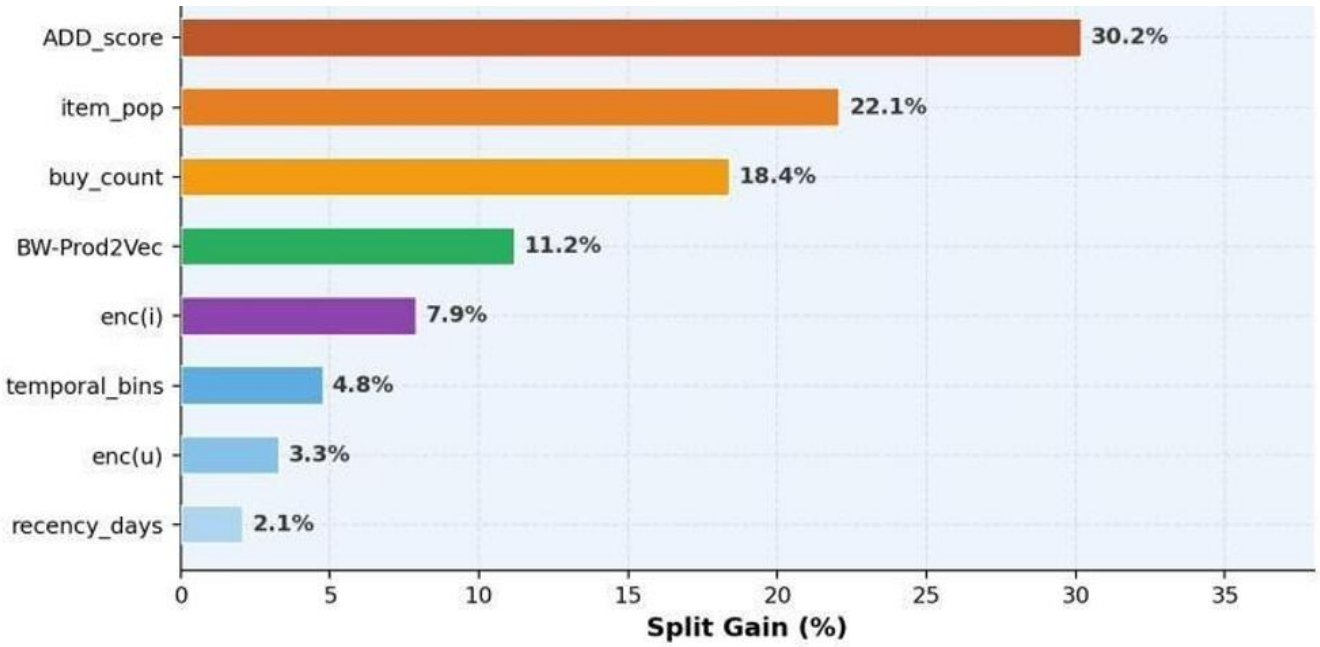

Fig. 6: Feature importance (split-gain). ADD_score leads at 30.2%; BW-Prod2Vec similarity at 11.2%.

7. CONCLUSION, LIMITATIONS, AND FUTURE WORK

We introduced two novel components for recommendation in time-sensitive e-commerce settings. Adaptive Dual-Decay (ADD) recency scoring replaces the conventional single-curve temporal decay with a user activity-aware blend that correctly handles both frequent and infrequent buyers. Behavior-Weighted Prod2Vec [7] extends standard item embeddings with graduated action-type weights that more faithfully reflect purchase intent. Together they improve Precision@10 by 12.1%

and Recall@10 by 16.4% over a strong LightGBM baseline, with the contribution of each component confirmed through ablation.

Several limitations deserve acknowledgment. The entire evaluation rests on a single 21-day promotional window from one platform [1], which limits how confidently the ADD tier boundaries and exponential rate generalize to other shopping environments with steadier demand. Our SASRec comparison relies on published numbers [9] rather than a controlled retraining, which is honest about the practical constraints but imperfect for direct benchmarking. Future work should examine whether the ADD parameterization transfers to non-promotional datasets and should benchmark against attention models under identical training budgets.

On the methodological side, the BW-Prod2Vec action weights are currently fixed by prior judgment; learning them jointly with the embedding stage perhaps via a lightweight attention layer would be a natural improvement. Including sessionlevel context (what the user browsed immediately before a recommendation request) is another unexplored direction that could yield further gains without substantially increasing system complexity.

Despite these limitations, both ADD and BW-Prod2Vec are practically straightforward: they can be layered onto any existing Prod2Vec–LightGBM stack with minimal code changes and negligible additional compute. We believe they offer a useful, interpretable path to better recommendation performance for teams who cannot justify the resource requirements of large sequential models.

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